

PE-EC702A — Adaptive Signal Processing

Semester VII • 3L:0T:0P • 3 Credits

Module 2 Exam Notes: Wiener Filters, Steepest Descent & LMS

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1. Optimal FIR (Wiener) Filter

The Wiener filter is the optimal linear filter in the Mean-Square Error (MSE) sense. It assumes the signals are wide-sense stationary with known second-order statistics (autocorrelation and cross-correlation).

1.1 Derivation of the Wiener-Hopf Equations

Let the input vector be $\mathbf{x}(n) = [x(n), x(n-1), \dots, x(n-M+1)]^T$ and the weight vector be $\mathbf{w} = [w_0, w_1, \dots, w_{M-1}]^T$. The filter output is $y(n) = \mathbf{w}^H \mathbf{x}(n)$. The error signal is $e(n) = d(n) - y(n) = d(n) - \mathbf{w}^H \mathbf{x}(n)$. The cost function to minimize is the Mean-Square Error $J(\mathbf{w}) = \mathbb{E}[|e(n)|^2]$:

$$J(\mathbf{w}) = \mathbb{E}[(d(n) - \mathbf{w}^H \mathbf{x}(n))(d^*(n) - \mathbf{x}^H(n) \mathbf{w})] \quad (1)$$

Expanding the expectation:

$$\begin{aligned} J(\mathbf{w}) &= \mathbb{E}[|d(n)|^2] - \mathbf{w}^H \mathbb{E}[\mathbf{x}(n) d^*(n)] - \mathbb{E}[d(n) \mathbf{x}^H(n)] \mathbf{w} + \mathbf{w}^H \mathbb{E}[\mathbf{x}(n) \mathbf{x}^H(n)] \mathbf{w} \\ &= \sigma_d^2 - \mathbf{w}^H \mathbf{p} - \mathbf{p}^H \mathbf{w} + \mathbf{w}^H \mathbf{R} \mathbf{w} \end{aligned} \quad (2)$$

where $\sigma_d^2 = \mathbb{E}[|d(n)|^2]$, $\mathbf{p} = \mathbb{E}[\mathbf{x}(n) d^*(n)]$, and $\mathbf{R} = \mathbb{E}[\mathbf{x}(n) \mathbf{x}^H(n)]$. To find the optimal weight vector $\mathbf{w}_o$, we differentiate $J(\mathbf{w})$ with respect to the complex conjugate of the weight vector $\mathbf{w}^*$ (or compute the gradient) and set it to zero:

$$\nabla_{\mathbf{w}^*} J(\mathbf{w}) = -\mathbf{p} + \mathbf{R} \mathbf{w} = \mathbf{0} \quad (3)$$

This yields the **Wiener-Hopf Equations**:

$$\mathbf{R} \mathbf{w}_o = \mathbf{p} \implies \mathbf{w}_o = \mathbf{R}^{-1} \mathbf{p} \quad (4)$$

1.2 Minimum Mean-Square Error ($J_{\min}$)

Substituting the optimal weight $\mathbf{w}_o$ back into the cost function $J(\mathbf{w})$ gives the minimum mean-square error:

$$J_{\min} = \sigma_d^2 - \mathbf{w}_o^H \mathbf{p} = \sigma_d^2 - \mathbf{p}^H \mathbf{R}^{-1} \mathbf{p} \quad (5)$$

2. Method of Steepest Descent

When the statistics $\mathbf{R}$ and $\mathbf{p}$ are known, but inverting $\mathbf{R}$ is computationally expensive, we use the iterative Method of Steepest Descent. The weight vector is updated in the direction opposite to the gradient of the performance surface $J(\mathbf{w})$:

2.1 Weight Update Equation

$$\mathbf{w}(n+1) = \mathbf{w}(n) - \frac{1}{2} \mu \nabla J(\mathbf{w}(n)) \quad (6)$$

where μ is the step-size parameter. Since $\nabla J(\mathbf{w}) = -2\mathbf{p} + 2\mathbf{R}\mathbf{w}(n)$, we have:

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \mu(\mathbf{p} - \mathbf{R}\mathbf{w}(n)) \quad (7)$$

2.2 Convergence Analysis and Stability

Let the weight error vector be defined as $\mathbf{c}(n) = \mathbf{w}(n) - \mathbf{w}_o$. Subtracting $\mathbf{w}_o$ from both sides of the update equation:

$$\mathbf{c}(n+1) = (\mathbf{I} - \mu\mathbf{R})\mathbf{c}(n) \quad (8)$$

Applying the spectral decomposition $\mathbf{R} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^H$ and defining the rotated error vector $\mathbf{v}(n) = \mathbf{Q}^H\mathbf{c}(n)$, we get:

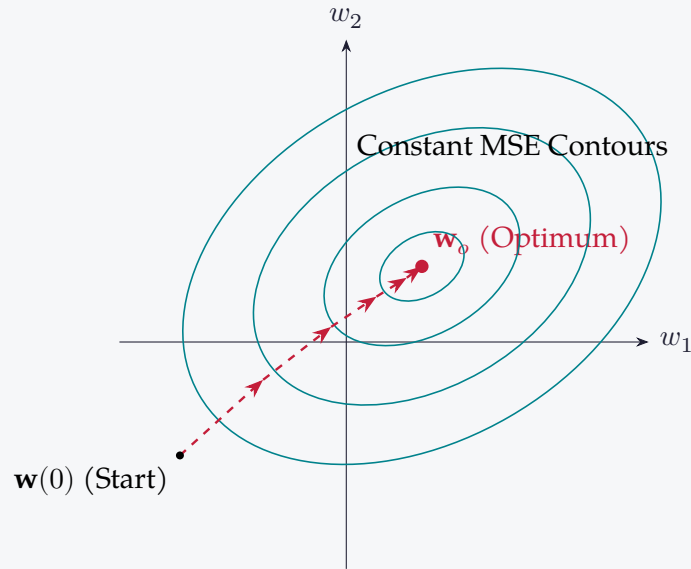
$$\mathbf{v}(n+1) = (\mathbf{I} - \mu\mathbf{\Lambda})\mathbf{v}(n) \quad (9)$$

For stability, each decoupled mode $v_i(n+1) = (1 - \mu\lambda_i)v_i(n)$ must decay to zero, which requires $|1 - \mu\lambda_i| < 1$ for all i :

$$0 < \mu < \frac{2}{\lambda_{\max}} \quad (10)$$

where $\lambda_{\max}$ is the largest eigenvalue of the correlation matrix $\mathbf{R}$.

Contour Lines of MSE Surface and Convergence Trajectory



3. The Least Mean Squares (LMS) Algorithm

The Least Mean Squares (LMS) algorithm is a stochastic gradient descent algorithm. It avoids the need to know or estimate the ensemble statistics $\mathbf{R}$ and $\mathbf{p}$ by using instantaneous approximations:

$$\hat{\mathbf{R}}(n) = \mathbf{x}(n)\mathbf{x}^H(n), \quad \hat{\mathbf{p}}(n) = \mathbf{x}(n)d^*(n) \quad (11)$$

3.1 LMS Algorithms Formulations

The instantaneous cost function is chosen as $\hat{J}(n) = |e(n)|^2$. The update equations are:

LMS Update Equations

Real-Valued LMS Algorithm:

$$e(n) = d(n) - \mathbf{w}^T(n)\mathbf{x}(n) \quad (12)$$

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \mu e(n)\mathbf{x}(n) \quad (13)$$

Complex-Valued LMS Algorithm:

$$e(n) = d(n) - \mathbf{w}^H(n)\mathbf{x}(n) \quad (14)$$

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \mu \mathbf{x}(n)e^*(n) \quad (15)$$

3.2 LMS Convergence Analysis

Under the **independence assumption** (the input vectors $\mathbf{x}(n)$ are statistically independent over time):

- **Convergence in the Mean:** The ensemble average of the weights converges to the Wiener solution, $\mathbb{E}[\mathbf{w}(n)] \rightarrow \mathbf{w}_o$, if:

$$0 < \mu < \frac{2}{\lambda_{\max}} \quad (16)$$

A conservative practical limit based on total signal power is:

$$0 < \mu < \frac{2}{\text{Tr}(\mathbf{R})} = \frac{2}{M \cdot R_x(0)} \quad (17)$$

- **Convergence in the Mean-Square:** For the second-order statistics of the weights to remain stable, the step-size must satisfy:

$$0 < \mu < \frac{2}{\sum_{i=1}^M \lambda_i} = \frac{2}{\text{Tr}(\mathbf{R})} \quad (18)$$

3.3 Excess MSE and Misadjustment

Due to gradient estimation noise, the LMS algorithm does not settle exactly at $J_{\min}$. It fluctuates about the optimum, introducing an **Excess Mean-Square Error** J_{ex} :

$$J_{\text{ex}} = \lim_{n \rightarrow \infty} \mathbb{E}[|e(n)|^2] - J_{\min} \quad (19)$$

For a small step-size μ , the excess MSE is given by:

$$J_{\text{ex}} \approx \frac{\mu J_{\min} \text{Tr}(\mathbf{R})}{2 - \mu \text{Tr}(\mathbf{R})} \quad (20)$$

The **Misadjustment** M is a dimensionless measure of how close the LMS filter is to the optimal Wiener filter:

$$M = \frac{J_{\text{ex}}}{J_{\min}} \approx \frac{\mu}{2} \text{Tr}(\mathbf{R}) = \frac{\mu}{2} \sum_{i=1}^M \lambda_i \quad (21)$$

Example: Wiener Filter Coefficient Design

A single-tap LMS filter has input signal $x(n)$ with variance $R_x(0) = 2$. The desired signal has variance $\sigma_d^2 = 5$, and the cross-correlation is $p = 1.6$.

1. Determine the optimal weight w_o .
2. Compute the minimum MSE $J_{\min}$.
3. Determine the maximum step-size μ for convergence in the mean.
4. Calculate the misadjustment M for $\mu = 0.05$.

Solution:

1. Using the Wiener-Hopf equation:

$$w_o = R^{-1}p = \frac{p}{R_x(0)} = \frac{1.6}{2} = 0.8$$

2. The minimum MSE $J_{\min}$ is:

$$J_{\min} = \sigma_d^2 - w_o p = 5 - (0.8 \times 1.6) = 5 - 1.28 = 3.72$$

3. For a single-tap filter ($M = 1$), $\lambda_{\max} = R_x(0) = 2$. The step-size range is:

$$0 < \mu < \frac{2}{\lambda_{\max}} \implies 0 < \mu < 1$$

4. The misadjustment M for $\mu = 0.05$:

$$M \approx \frac{\mu}{2} \text{Tr}(\mathbf{R}) = \frac{0.05}{2} \times 2 = 0.05 \text{ (or 5\%)}$$

Quick Revision — Module II

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Key Formulas at a Glance

Wiener-Hopf Equation	$\mathbf{R}\mathbf{w}_o = \mathbf{p} \Rightarrow \mathbf{w}_o = \mathbf{R}^{-1}\mathbf{p}$
Minimum MSE	$J_{\min} = \sigma_d^2 - \mathbf{p}^H \mathbf{w}_o$
Steepest Descent Update	$\mathbf{w}(n+1) = \mathbf{w}(n) + \mu(\mathbf{p} - \mathbf{R}\mathbf{w}(n))$
LMS Update (Complex)	$\mathbf{w}(n+1) = \mathbf{w}(n) + \mu \mathbf{x}(n) e^*(n)$
Step-Size Bounds	$0 < \mu < \frac{2}{\lambda_{\max}}$ (Ensemble) or $0 < \mu < \frac{2}{\text{Tr}(\mathbf{R})}$ (LMS)
Misadjustment	$M = \frac{J_{\text{ex}}}{J_{\min}} \approx \frac{\mu}{2} \text{Tr}(\mathbf{R})$

One-Line Definitions

- **Wiener Filter:** The optimal linear time-invariant filter that minimizes mean-square error.
- **Steepest Descent:** An iterative optimization method that moves weights along the direction of maximum negative gradient.
- **LMS Algorithm:** A low-complexity adaptive algorithm using instantaneous signal values to approximate gradients.
- **Misadjustment:** A measure of the average excess MSE relative to the minimum MSE, representing algorithm noise.
- **Eigenvalue Spread:** The ratio $\lambda_{\max}/\lambda_{\min}$, indicating the convergence rate limit of gradient algorithms.
- **Independence Assumption:** An analytical simplification stating that successive tap-input vectors are statistically independent.

Frequently Asked / Viva Questions

Q1. Derive the Wiener-Hopf equations for a real FIR filter.

See Section 1.1: We expand $J(\mathbf{w}) = \mathbb{E}[(d(n) - \mathbf{w}^T \mathbf{x}(n))^2] = \sigma_d^2 - 2\mathbf{w}^T \mathbf{p} + \mathbf{w}^T \mathbf{R} \mathbf{w}$. Differentiating w.r.t. $\mathbf{w}$ and setting to $\mathbf{0}$ yields $\mathbf{R}\mathbf{w}_o = \mathbf{p}$.

Q2. What is the convergence criterion for the method of steepest descent?

The step-size parameter must satisfy $0 < \mu < 2/\lambda_{\max}$, where $\lambda_{\max}$ is the maximum eigenvalue of the autocorrelation matrix $\mathbf{R}$.

Q3. Why is LMS referred to as a "stochastic" gradient algorithm?

Because it replaces the true gradient (which requires expectations of signals) with a noisy instantaneous gradient estimate computed directly from the current

Frequently Asked / Viva Questions (contd.)

samples.

Q4. Write down the complex LMS weight update equation and explain each term.

$\mathbf{w}(n+1) = \mathbf{w}(n) + \mu \mathbf{x}(n) e^*(n)$, where $\mathbf{w}(n)$ is the weight vector, μ is the step-size, $\mathbf{x}(n)$ is the input vector, and $e^*(n)$ is the complex conjugate of the error.

Q5. What is the eigenvalue spread and how does it affect convergence speed?

Eigenvalue spread is the ratio $\chi = \lambda_{\max}/\lambda_{\min}$. A large eigenvalue spread (highly correlated input) causes the MSE surface to form a narrow valley, resulting in slow convergence.

Q6. What is the physical meaning of misadjustment in LMS?

It represents the cost of adaptation; the weight vector continuously hunts around the optimum, causing the average MSE at steady state to be slightly larger than $J_{\min}$.

Q7. What is the conservative step-size limit used in practice for LMS?

$\mu < 2/\text{Tr}(\mathbf{R}) = 2/(M \cdot \text{Signal Power})$, which is easily calculated since it only requires the input signal power.

Q8. Can the LMS algorithm converge if the step-size μ is negative?

No, if $\mu < 0$, the algorithm moves in the direction of the positive gradient, leading to divergence (instability) and weights going to infinity.

Q9. How does the number of filter taps M affect the misadjustment?

Since $M \approx \text{Tr}(\mathbf{R})/P_x$, a larger number of filter taps increases $\text{Tr}(\mathbf{R})$, which increases the misadjustment for a fixed step-size μ .

Q10. What is the minimum MSE ($J_{\min}$) of a system where the input and desired signal are uncorrelated?

If they are uncorrelated, the cross-correlation vector $\mathbf{p} = \mathbf{0}$, giving $\mathbf{w}_o = \mathbf{0}$, which yields $J_{\min} = \sigma_d^2$ (the total power of the desired signal).